MTH209: Midsem Exam - Question 2

1. Follow the instructions **exactly**.
2. If your code is not properly commented or horizontal/vertical spacing is missing, points will be deducted.
3. You are only allowed to use the libraries that we have discussed in the course, and no other libraries are allowed to be called.

Question

The goal of this problem is to compare two estimators of the shape parameter of a Gamma distribution. Recall that a Gamma$(\alpha, 1)$ distribution has mean α and variance α. Suppose $X_1, X_2, \dots, X_n \overset{iid}{\sim}$ Gamma$(\alpha, 1)$. Then we have two estimators of α:

1. $T_1 = \dfrac{1}{n} \sum_{i=1}^{n} X_i$
2. $T_2 = \dfrac{1}{n-1} \sum_{i=1}^{n} (X_i - T_1)^2$

The goal is to write a function, `comp_alpha`, that takes in the value of α, the values of n for comparison, and the required numbers of replications, and makes a plot of the MSE of the two estimators with appropriate labels and legends. Here is what the function will look like:

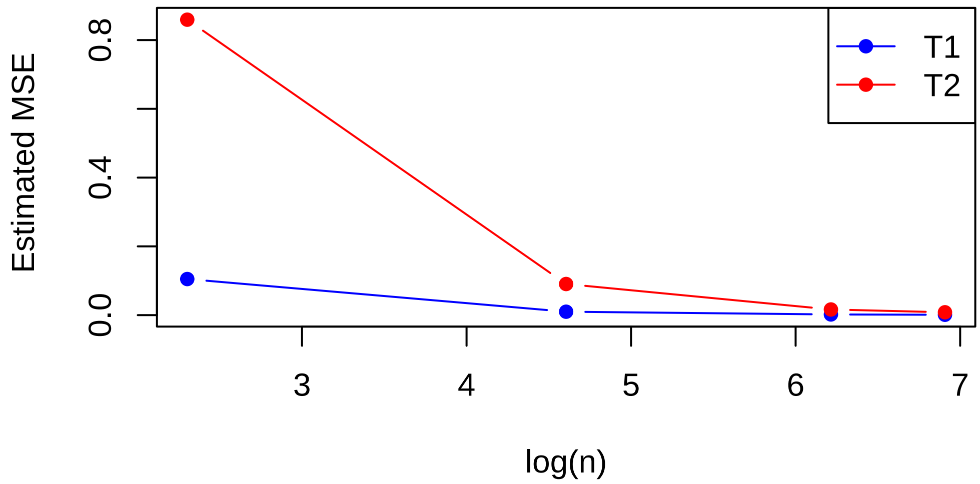
```
#|eval: false
# alpha = shape parameter
# n.vec = grid of n values to calculate the MSE for
# reps = replications
comp_alpha <- function(alpha, n.vec = c(10, 100, 500, 1000), reps)
{
  ...

  # plot the MSE with log(n) on the x-axis
  plot(log(n.vec), ...)
  # legend
  ...
}
```

Testing

The following is the expected output when I run this line. There will be some variability due to randomness.

```
comp_alpha(alpha = 1, reps = 1000)
```



Submission Instructions

INSTRUCTIONS: copy and paste the final function into a file called `question2.R` and upload it to mooKIT. Upload **ONLY the function** and nothing else. This question has 25 marks.